ARC243-HW01-Dash Altankhuyag

PART #:01

Electrical usage / BTUs

Total BTU	764	kwh	x	3412	btu/KW	=	2,606,768.00	BTU
Total COST	$104.76	dollar	+	192.76	given	=	$297.52	dollar
Cost/MMBTU	$297.52	total cost	/	7.766	MMBTU	=	$38.31	MMBTU

Natural Gas / BTUs

Total BTU	27	CF	x	1030	BTU /CF	=	27810	BTUs
Total COST	31.1							
Cost/MMBTU	31.1		/	12.875	MMBTU	=	2.415533981	MMBTU

Gasoline /BTUs

Weekday	20	miles/day	x	5	day/week	x	4.3	week/month	=	430	miles
Weekend	45	miles/day	x	2	day/week	x	5.3	week/month	=	477	miles
Vehicular performance	22	miles/per Gallon	@	3.66	per gallon						

Total BTU	907	miles	/	22	M/Gal	x	125000	BTU /Gal	=	5,153,409.09	BTUs
Total Cost:	40	Gal	x	$ 3.66	Gal	=	$ 146.40				
Cost/MMBTU	146.4	total cost	/	9.268	MMBTU	=	15.7962883	MMBTU			

Serivce	MMBTU		$/Month		$/MMBTU	
Electricity	7.766	25.97%	$297.52	62.63%	$38.31	68%
Natural Ga	12.875	43.05%	31.1	6.55%	2.41553398	4%
Gasoline	9.268	30.99%	$ 146.40	30.82%	15.7962883	28%
Monthly	29.909	100.00%	$475.02	100.00%	$56.52	100%

What is your choice of fuel source for your building? “Natural Gas”

I agree that natural gas is a good choice of fuel source for buildings. It is relatively inexpensive, clean-burning, and abundant in the United States. Natural gas is also a versatile fuel that can be used for a variety of purposes, including heating, and cooking.

Here are some of the reasons why natural gas is a good choice of fuel source for my buildings:

- Cost: Natural gas is typically less expensive than electricity, especially for heating purposes.
- Cleanliness: Natural gas burns more cleanly than other fossil fuels.
- Abundance: The United States has abundant reserves of natural gas. This means that it is a reliable and secure source of energy.

- Natural gas can be used for a variety of purposes, including heating, cooking, and electricity generation. This makes it a versatile and flexible fuel source.

It is a relatively inexpensive, clean-burning, and abundant fuel that can be used for a variety of purposes.

PART #:02

<u>Thermal Conductivity</u>

To convert thermal conductivity from watts per meter per degree Celsius (W/m/°C) to British thermal units per hour per square foot per degree Fahrenheit (BTU/h/ft²/°F), use the following formula:

$BTU/h/ft^2/°F = W/m/°C * 6.9334713$

if the thermal conductivity is 2.0 W/m/°C, then the equivalent in $BTU/h/ft^2/°F$ is:

$BTU/h/ft^2/°F = 2.0 W/m/°C * 6.9334713 = 13.8669426 BTU/h/ft^2/°F$

Therefore, 2.0 W/m/°C is equivalent to 13.8669426 $BTU/h/ft^2/°F$.

<u>Temperature:</u>

To convert temperature from Celsius (C) to Fahrenheit (F), you can use the following formula:

$F = C * 9/5 + 32$

So, if the temperature is 20 C, then the equivalent in F is:

$F = 20 * 9/5 + 32 = 68°F$

Therefore, 20 C is equivalent to 68°F.

Note:

Fahrenheit. adjective. Fahr·en·heit ˈfar-ən-ˌhīt. : relating or conforming to or having a thermometer scale on which under standard atmospheric pressure the boiling point of water is at 212 degrees above the zero of the scale and the freezing point is at 32 degrees above zero. abbreviation F.

Heat Flux

To convert heat flux from watts per meter (W/m) to British thermal units per hour per square foot ($BTU/h/ft^2$), you can use the following formula:

$BTU/h/ft^2 = W/m * 3.4129411733333335 * 1/ft^2$

So, if the heat flux is 1000 W/m, then the equivalent in $BTU/h/ft^2$ is:

$BTU/h/ft^2 = 1000\ W/m * 3.4129411733333335 * 1/ft^2$
$= 3412.941173333335\ BTU/h/ft^2$

Therefore, 1000 W/m is equivalent to 3412.941173333335 $BTU/h/ft^2$.

Temperature Difference

A 20-degree Celsius difference is equivalent to a 68 degree Fahrenheit difference.

To convert a temperature difference from Celsius (C) to Fahrenheit (F), you can use the following formula:

F = C * 9/5 + 32

So, if the temperature difference is 20 C, then the equivalent in F is:

F = 20 * 9/5 + 32 = 68°F

Therefore, a 20-degree Celsius difference is equivalent to a 68-degree Fahrenheit difference.

PART #:03

Conductivity, conductance, and thermal resistance are all terms used to describe the ability of a material to transfer heat.

Conductivity is a material property that measures how well heat can flow through the material. It is measured in watts per meter per kelvin (W/m/K).

Conductance is the amount of heat that can flow through a material per unit time per unit temperature difference. It is calculated by multiplying the conductivity of the material by the area of the cross-section and the length of the material. It is measured in watts per kelvin (W/K).

Thermal resistance is the inverse of conductance. It is a measure of how well a material resists the flow of heat. It is measured in kelvin per watt (K/W).

In other words, conductivity is a property of the material, while conductance and thermal resistance are properties of the material and the geometry of the system.

Here is a table that summarizes the differences between conductivity, conductance, and thermal resistance:

Conductivity	Measures how well heat can flow through a material	Watts per meter per kelvin (W/m/K)
Conductance	Measures the amount of heat that can flow through a material per unit of time per unit of temperature difference	Watts per kelvin (W/K)
Thermal resistance	Measures how well a material resists the flow of heat	Kelvin per watt (K/W)

Here are some examples of how conductivity, conductance, and thermal resistance are used in everyday life:

- Conductivity: The conductivity of a material is important for a variety of applications, such as cooking, electronics, and insulation.
- Conductance: The conductance of a material is important for heat transfer applications, such as cooling systems and heating systems.
- Thermal resistance: The thermal resistance of a material is important for insulation applications. For example, a thermal blanket has a low thermal resistance, so it can prevent heat from escaping from the body. A house insulation has a low thermal resistance, so it can prevent heat from escaping from the house.